

Soumyadip Dutta

soumyadipdutta2004@gmail.com | 9830351673 | [Portfolio](#) | [LinkedIn](#) | [GitHub](#)

Education

National Institute of Technology, Durgapur

Aug 2023 – Present

Bachelor of Technology in Electrical Engineering

CGPA: 9.23/10

Baranagore Ramakrishna Mission Ashrama High School (H.S.) 2011 – 2023

Class 1–10: 90.85%

Class 11–12: 94.00%

Experience

Summer Research Intern, IIT BHU

May 2025 – June 2025

Simulation and Optimization of High Voltage Insulation in High Temperature Superconducting (HTS) Cables

- Completed a 7-week research internship under Prof. Arup Kumar Das.
- Simulated electric field distribution in High Temperature Superconducting (HTS) cables using COMSOL.
- Analyzed effects of butt gaps, voids, stress cone thickness, and insulation thickness on field stress.
- Conducted partial discharge testing and thermal aging analysis on Kapton and PTFE insulation and optimized insulation parameters to improve reliability of high voltage HTS cable systems.
- **Tools:** COMSOL Multiphysics, Python.

Skills

HDL & Digital Design:	Verilog, RTL Design, FSM Design
EDA Tools:	Vivado, KiCad, LTSpice, Multisim, Icarus Verilog
Programming Languages:	Python, C, MATLAB
Simulation:	COMSOL Multiphysics, Simulink
Embedded Systems:	Arduino, ESP32-CAM
Languages:	English, Hindi, Bengali
Interests:	Embedded Systems, IoT, Robotics, Machine Learning

Projects

- **MIPS32 5-Stage Pipelined Processor in Verilog**
 - Designed and implemented a 32-bit MIPS processor using a 5-stage pipelined architecture.
 - Developed core modules and implemented pipeline registers, data forwarding and hazard detection logic to improve execution efficiency.
 - Verified processor functionality through simulation and custom testbenches.
 - **Tools:** Verilog HDL, Icarus Verilog, GTKWave.
 - **GitHub:** [\[Link\]](#)
- **SVPWM Controlled Three-Phase Inverter**
 - Designed and simulated a Space Vector Pulse Width Modulation (SVPWM) algorithm for a three-phase voltage source inverter.
 - Implemented Clarke transformation, sector detection, and switching time calculation (T1, T2, T0).
 - Generated PWM gate signals for a six-switch MOSFET inverter using carrier comparison.
 - Verified inverter operation by driving a star-connected RL load and observing phase voltage waveforms.
 - **Tools:** Simulink.

- **GitHub:** [\[Link\]](#)
- **Two-Product Vending Machine on Spartan-7 FPGA**
 - Designed and implemented an RTL-based vending machine controller on Spartan-7 FPGA using Verilog HDL.
 - Developed FSM-based control logic with modular datapath and control unit architecture.
 - Verified functionality using Verilog testbenches and performed synthesis using Xilinx Vivado.
 - Deployed on FPGA hardware with real-time I/O interfacing.
 - **Tools:** Verilog HDL, Xilinx Vivado, Spartan-7 FPGA.
 - **GitHub:** [\[Link\]](#)
- **ESP32-CAM Based QR Code Scanner**
 - Developed a wireless QR scanner using ESP32-CAM.
 - Streamed video from ESP32-CAM over WiFi and implemented real-time QR code detection using OpenCV and Pyzbar.
 - **Tools:** ESP32-CAM, Arduino IDE, OpenCV, NumPy, Pyzbar.
 - **GitHub:** [\[Link\]](#)
- **Arduino Uno PCB Design**
 - Designed a custom Arduino Uno PCB using KiCad.
 - Implemented schematic design, component selection and PCB layout optimization.
 - **Tools:** KiCad.
 - **GitHub:** [\[Link\]](#)

Relevant Courses

Analog Electronics, Digital Electronics, Digital Signal Processing, Control Systems, Power Electronics, Electrical Machines, Microprocessor and Microcontroller, FPGA based Design, Advanced Power Electronics

Academic Achievements

- GATE EE-2026 AIR-2309 (Score - 507)
- JEE Main CRL: 16327 (98.60 percentile)
- Rank 15 in PUBDET (Presidency University Entrance Exam) - Mathematics
- Reached National Level (2nd Round) of NTSE Exam - 2021